BTS–IV(R/S)–06–24–2785 Reg. No. 20222024

A

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

CS 19-202-0403 COMPUTER ARCHITECTURE AND ORGANIZATION

(2019 Scheme)

Time: 3 Hours Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Acquire knowledge about structure, functions and characteristics of computer systems.
CO2: Identify the addressing modes used in instructions.
CO3: Determine the set of control signals generated and their timing sequence, given an instruction.
CO4: Demonstrate how addition, multiplication and division operations are implemented inside a computer system.
CO5: Explain each level of memory hierarchy.
CO6: Show how cache mapping affect the location of the data and the replacement policies.
CO7: Map a virtual address to physical address.
CO8: Identify and compare different methods for computer I/O.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A
(Answer *ALL* questions)

		$(8 \times 3 = 24)$	Marks	BL	CO	PO
I.	(a)	Distinguish two address and three address instructions.	3	L2	2	1,2
	(b)	Discuss the single bus organization with a neat diagram.	3	L2	1	1,2
	(c)	Illustrate how fetching a word from memory is done using control signals.	3	L1	1	1
	(d)	Discuss hardwired control.	3	L2	3	1,2,4
	(e)	Explain fast page mode in DRAM.	3	L2	5	2
	(f)	Discuss the terms cache hit, cache miss and miss penalty.	3	L2	5	1,3
	(g)	Discuss how multiple interrupts from a single device is handled.	3	L1	8	1,3
	(h)	Explain distributed bus arbitration.	3	L2	8	1,3

PART B

		$(4 \times 12 = 48)$	Marks	BL	CO	PO
II.	(a)	Explain the various addressing modes with suitable examples.	6	L2	2	1
	(b)	Discuss subroutines. Also explain subroutine nesting and parameter passing.	6	L1	1	1
		OR				
III.	(a)	Describe various operations in stack with appropriate diagrams.	6	L1	1	1
	(b)	Discuss the types of instructions with examples.	6	L2	2	1
IV.	(a)	Explain Booth's algorithm. Compute 14*-7 using the algorithm.	6	L3	4	1,3,4
	(b)	Explain n bit ripple carry adder and its delay calculation.	6	L1	4	1,3,4
		OR				
V.	(a)	Illustrate a 4-bit carry save look ahead adder and derive the expressions for carry signals.	6	L3	4	1,3,4
	(b)	Explain the principle of restoring division with an example.	6	L1	4	1,3,4

(P.T.O.)

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			Marks	BL	CO	PO
VI.	(a)	Explain any two mapping functions used in cache memory with physical address formats.	7	L2	6	1,3,4
	(b)	Discuss address translation using page table with a neat diagram.	5	L2	7	1,3,4
		OR				
VII.	(a)	What is the average memory access time for a machine with a cache hit rate of 80% and cache access time of 5 ns and main memory access time of 100 ns when- (i) Simultaneous access memory organization is used (ii) Hierarchical access memory organization is used.	5	L2	6	1,3,4
	(b)	Describe virtual memory and address translation using TLB.	7	L2	7	1,3,4
VIII	(a)	Explain the working of serial ports.	8	L1	8	1,3
	(b)	Discuss priority arbitration and daisy chaining in handling interrupt requests.	4	L2	8	1,3
		OR				
IX.	(a)	Explain the detailed operation on DMA.	8	L2	8	1,3
	(b)	Describe SCSI bus.	4	L1	8	1,3

Blooms's Taxonomy Level's
L1 – 33.3%, L2 –58.33%, L3 – 8.3%.

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